

Digital Signal Analysis and Processing (CT704)

Concise PYQ

July 22, 2026

Notes: TU IOE PYQ 2069–2081 (CT704, BEI/BCT). Regular = **bold**, Back = normal. Marks in (); unique values only. Keys: (I)=Importance/Need (+)=Advantage (-)=Disadvantage +Fig=Figure +Eg=Example Vs=Compare/Differentiate.

1. Discrete-Time Signals & Systems

- Even/odd DT signals +Eg; even/odd part; plot $x[-2n+3]$ +Fig (2+3)(3)(4+5)
- Unit step +Fig; relation to unit impulse (1+2)
- Energy Vs power signal; check type (2+2)(2+3)(3+4)
- Periodicity + fundamental period (5)(4)(3)(2+2)
- System properties: linear / time-invariant / stable / memoryless; BIBO (2+2+2)(3+3)(5)
- Fourier series coeff; FT multiplication property; Dirichlet conditions (3)(4)(4+3)
- LTI convolution output (various h, x) (4)(5)(6)(3+7)
- Significance of convolution summation + compute (2+4)

2. Z-Transform

- ROC definition + properties + locate ROC (1)(2+4)(3+6)(4+6)
- Z-transform properties: convolution, time-shift (3+3)(3+6)
- Inverse Z-transform (partial fraction / long division), various $X(z)$ & ROC (6)(1+5)(2+6)(2+2+3)

3. Analysis of LTI in Frequency Domain

- Pole-zero + magnitude / phase response (various diff. eqns) (2+4)(3+7)(2+6)(2+8)
- FIR Vs IIR system (4+6)
- LCCDE + system function; zero-input response (3+7)(2+3+5)(4)
- Stability & causality via impulse response & ROC (4)(4+3)
- Linear-phase FIR condition; show $h = \{-1, 0, 1\}$ linear phase (2+4)

4. Discrete Filter Structures

- Direct Form I & II (various systems) (2+2)(4)(5)
- Cascade form, 2nd-order sections (4)(5)
- Lattice / lattice-ladder (FIR & IIR) + stability (5)(6)(6+4)(9)(10)

- 3-stage lattice $K_1, K_2, K_3 \rightarrow$ system function + FIR coeff (5)(6)
- represent fraction in sign-mag / 1's / 2's complement (3)
- Limit cycle in recursive system +Eg (1)(3)
- Quantization (I); rounding Vs truncation; dead band (1+1+2+1)

5. FIR Filter Design

- Window method + window characteristics; Gibbs phenomenon (rect-angular) + minimize (4+4)(5+3)(3+3)(5)(6)
- FIR LPF via window (specs $\omega_p, \omega_s, \alpha_s$; Hanning; first coeffs) (8)(10)(5+3)(2+6)(9)
- Kaiser window design (ripple / β / length $M+1$) (8)(10)(2+8)(9+3)(8+4)
- Optimum filter / Remez exchange (math expr + flowchart + derivation) (1+6)(2+6)(7)(9)

6. IIR Filter Design

- Bilinear-transformation Butterworth LPF (various specs); LP $\rightarrow$ HP convert (10)(12)(15)(11+4)
- Impulse-invariance Butterworth; pre-warping (I) (12+3)(13)
- Bilinear Vs impulse invariance (+) (2+10)(3+12)
- Spectral (digital-domain) transformation LP $\rightarrow$ HP; features / params (2)(4)(5)

7. Discrete Fourier Transform (DFT / FFT)

- DFT need; DFT Vs DTFT; complexity DFT Vs FFT (2)(2+5)
- Circular convolution property (state + prove); zero padding (2+2+4)(1)
- FFT (DIT / DIF) N -pt DFT + butterfly (many sequences) (7)(8)(2+6)(3+7)
- Circular convolution — direct / via DFT (6)(7)(2+5)(1+5)
- $x_3 = \text{IDFT}(X_1 \cdot X_2)$ (product of DFTs) (5)(6)(7)(2+6)